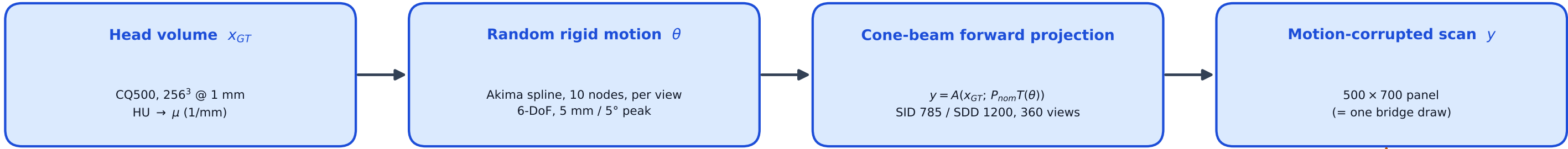


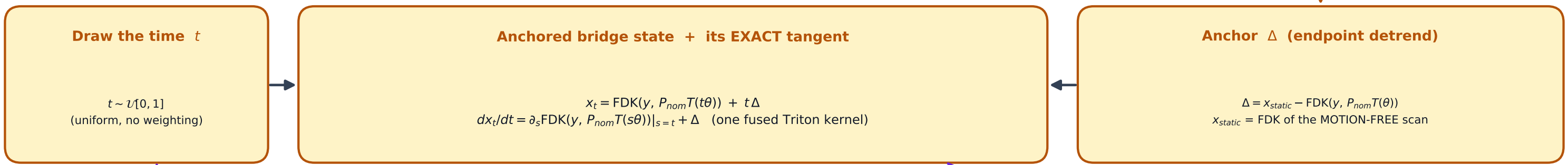
Training: a flow-matching prior on the GEOMETRY BRIDGE

the prior is trained on exactly the image manifold the inference ODE walks: reconstructions under a PARTIALLY CORRECTED geometry

1. Simulate a motion-corrupted scan (CQ500, train split)



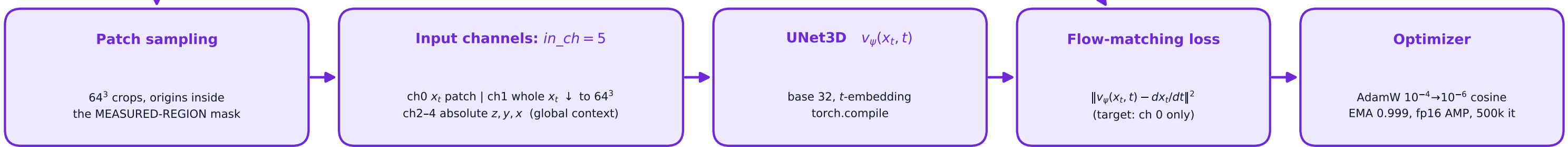
2. Geometry bridge (whole volume, no_grad — one fused FDK+tangent pass)



$t = 0$: $x_0 = \text{FDK}(y, P_{nom})$ — the uncorrected recon, which is exactly the INFERENCE COLD START

$t = 1$: $x_1 = x_{static}$ — an image this scanner can really produce (the bare bridge endpoint $\text{FDK}(\theta_{true})$ is still 1-3 dB short)

3. Train the patch prior (the net only ever sees 64^3 patches)



MEMORY SPLIT — the OPERATOR (FDK + its tangent) runs on the full volume under no_grad; the NET only ever sees 64^3 patches. That is what lets a 3D prior train on one 24 GB card.

A rolling cache of 8 whole-volume bridge draws is refreshed every 12 steps, and each batch of 8 patches mixes several draws, so t varies within a batch. Validation = prior-only ODE from the cold start on held-out patients: 25.42 dB / SSIM 0.78 (rigid-aligned).